Ethical vs. Deceptive Visualization: Design Rationale and Reflection

Overview of the Process

For this assignment, I worked with the **City of Seattle Wage Data** dataset to explore variations in hourly wages across different departments. The dataset was chosen because it provides transparency into public salaries and offers opportunities for ethical and misleading data representation.

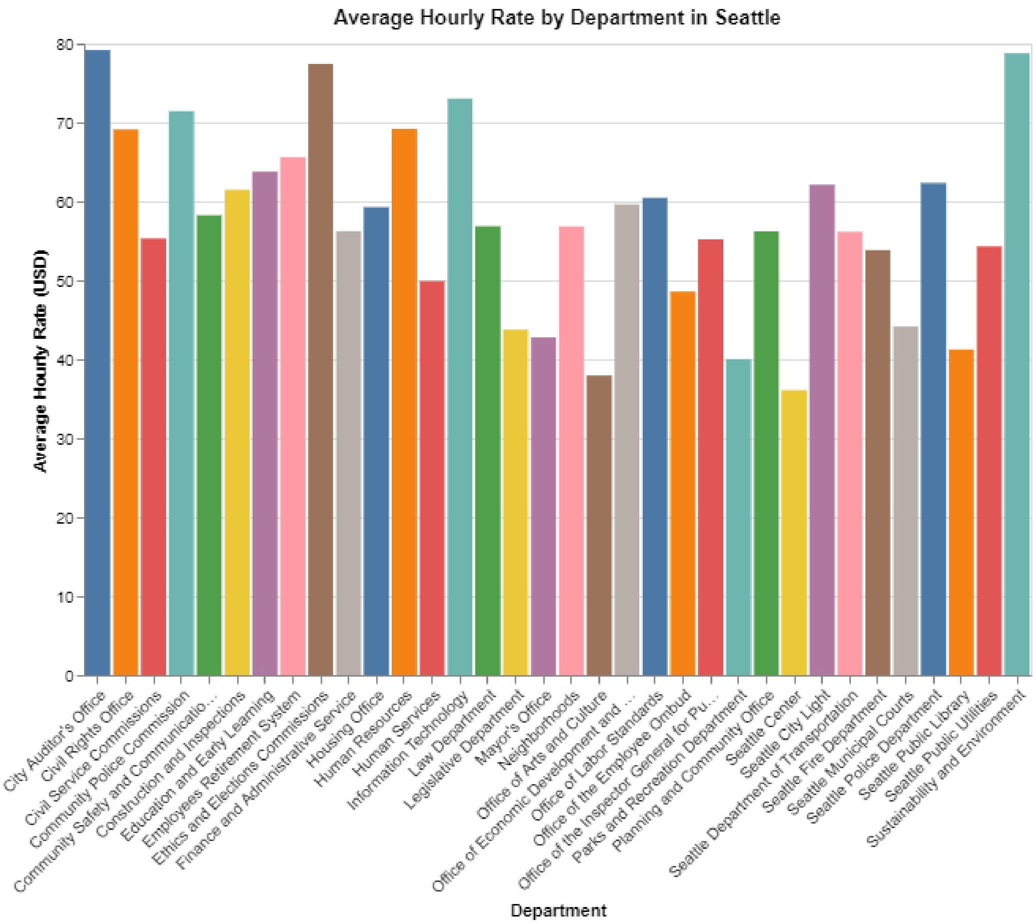
I formulated the key question: *How do hourly wages compare across different Seattle city departments?* This question is relevant because it directly impacts public perception of wage distribution and fairness. To address this, I created two visualizations: one ethical and one deceptive. My approach involved selecting appropriate encodings, graphical marks, and aggregation methods to ensure clarity in the ethical visualization while deliberately introducing misleading elements in the deceptive version.

Ethical Visualization

For the ethical visualization, I used a **bar chart** to show the **average hourly wage** by department. The **x-axis** represents departments, and the **y-axis** represents the mean hourly wage. I used consistent color encoding to avoid introducing unnecessary bias and provided clear axis labels and a descriptive title.

This visualization effectively answers the question by providing an accurate representation of wage distribution across departments. Aggregating by mean ensures that outliers do not disproportionately influence the perception of wage differences. The clear encoding of numerical values and department names allows viewers to easily compare different departments.

Screenshot:



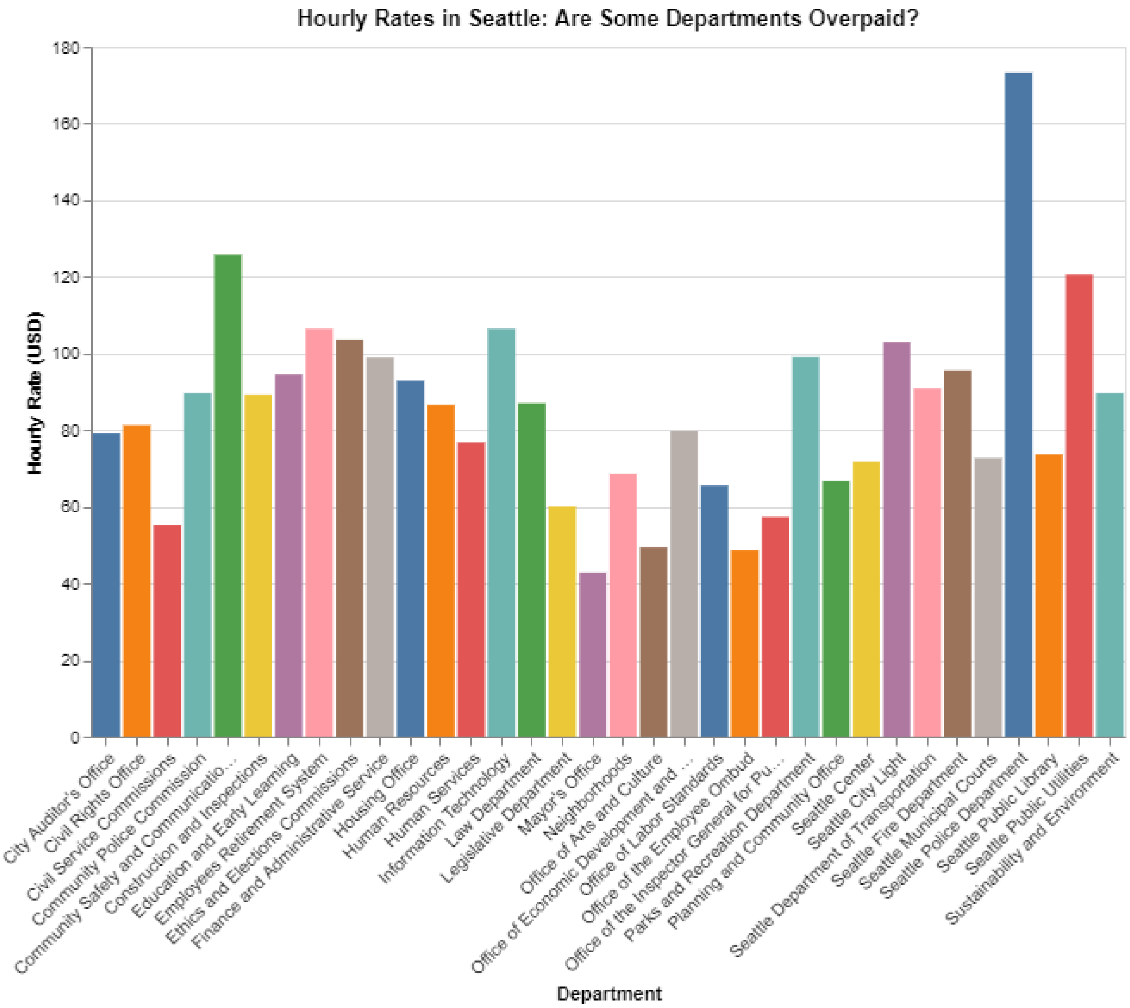
Observable Notebook Link:https://observablehq.com/d/322c2c0caea81096

Deceptive Visualization

For the deceptive visualization, I also used a **bar chart** but manipulated key design elements to mislead viewers. Instead of using the **mean** hourly wage, I aggregated the data by **maximum hourly wage**, making it appear as though all employees in a department earn at the highest reported rate. Additionally, I **truncated the y-axis**, exaggerating the differences between departments.

These choices mislead the viewer by overemphasizing certain wage disparities. A casual observer might mistakenly assume that departments with higher maximum salaries are overpaid, even though these values represent only a small subset of employees. The absence of proper context (such as median wage) further contributes to a distorted interpretation.

Screenshot:



Observable Notebook Link:https://observablehq.com/d/feb236c1fcf5d8ed

Learning Reflection

This assignment reinforced the importance of ethical data visualization and how small design choices can significantly impact viewer interpretation. I learned that aggregation methods, axis scaling, and color encoding play a crucial role in conveying an accurate message.

One of the biggest challenges was designing a misleading visualization that still appeared plausible. It required understanding how viewers perceive graphical representations and strategically applying distortions. This exercise also made me more critical of data visualizations in media, as I now recognize common techniques used to manipulate public perception. Overall, this project deepened my appreciation for responsible data communication and the ethical implications of visualization design.